

Adrian A. Charles

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EDUCATION

Queen's University

Bachelor of Applied Science in Computer Engineering

AWS Certified Data Engineer – Associate (in progress)

Kingston, Ontario

May 2026

EXPERIENCE

GE Vernova

Software Engineering Intern

Peterborough, Ontario

May 2024 – Aug 2024

- Designed and developed a PHP-based report generation tool used by 5 senior field engineers, reducing manual report generation time by 50% and enabling structured database storage for historical querying of previously unstructured Word documents.
- Built and deployed a web UI (HTML, CSS) allowing engineers to generate standardized reports on demand, eliminating manual formatting overhead across recurring reporting workflows.
- Maintained and upgraded existing internal tools through CI/CD pipelines, resolving bugs and shipping feature improvements while demonstrating end-to-end ownership of deployed systems.

GE Vernova

Software Engineering Intern

Peterborough, Ontario

May 2023 – Aug 2023

- Developed full-stack internal applications (PHP, MySQL, JavaScript) including a Customer Service Dashboard and Document Locator Tool, improving document retrieval speed by 10x for 100+ users by replacing a manual cabinet-based search process.
- Designed a 2-table relational data model and implemented SQL queries supporting 1,000+ records, ensuring structured data accessibility across engineering workflows.

Queen's University – Faculty of Engineering and Applied Science

Teaching Assistant - Intro to Computer Programming

Kingston, Ontario

Sept 2023 – Dec 2023

- Supported 800+ first-year engineering students in C programming, debugging, and algorithmic problem solving across weekly lab and office hour sessions.
- Graded 40+ assignments per term and delivered technical feedback on complex C concepts to beginner students, maintaining consistent turnaround within a 10-hour marking cycle.
- Balanced full academic courseload alongside instructional responsibilities, demonstrating time management and delivery under deadline.

PROJECTS

Cancer Clinical Trials Data Pipeline – Data Engineering Project

Developer

Peterborough, Ontario

June 2026

- Built an end-to-end ELT pipeline ingesting 100+ cancer clinical trial records daily from the ClinicalTrials.gov REST API, reducing manual data retrieval to zero by orchestrating automated DAGs with Apache Airflow on Docker Compose.
- Achieved full data quality coverage across critical fields (trial ID, status, title) by implementing automated dbt tests validating uniqueness and null constraints on every pipeline run.
- Accelerated analytical query performance by loading and transforming raw JSON into a structured BigQuery warehouse using a truncate-and-reload pattern, with dbt staging and mart models surfacing enrollment trends across 8 trial status categories and 70+ unique sponsors.
- Delivered stakeholder-facing insights through a Streamlit dashboard visualizing trial counts, sponsor activity, and enrollment date ranges sourced directly from dbt mart models.

RAG-Based AI Coding Assistant – VS Code Extension

Developer

Kingston, Ontario

September 2025 – April 2026

- Achieved 85% Top-1 bug localization accuracy and 75% patch success rate across 400 evaluations by building a RAG pipeline on PostgreSQL + pgvector with cosine similarity search, more than doubling a no-retrieval baseline (40% to 85%).
- Enforced data quality at every pipeline stage — ingestion, chunking, embedding, and retrieval — by building validation checks that flagged inconsistent outputs before they reached the reporting layer, keeping accuracy stable across 400 evaluations.

AWS Cloud Engineer Agent – AI Cloud Infrastructure Assistant

Developer

Kingston, Ontario

February 2026

- Developed an intelligent AWS Cloud Engineer agent using Python, Strands Agents SDK, and Amazon Bedrock, enabling natural language interaction for inspecting and reasoning about AWS infrastructure.
- Integrated a general-purpose boto3 wrapper via the Strands Agents SDK, enabling natural language interaction with 300+ AWS services including EC2, S3, CloudWatch, and IAM for automated infrastructure inspection and operational insights.

- Built a Streamlit-based conversational interface and incorporated external tools for AWS documentation retrieval and architecture diagram generation.

NBA Analytics Pipeline — Kafka, Airflow, Databricks (PySpark/Delta Lake), Streamlit

Kingston, Ontario

Developer

November 2025

- Engineered a real-time streaming ingestion pipeline processing 2,460+ NBA game records via a Kafka producer/consumer architecture, eliminating manual data pulls by containerizing both services with Docker and orchestrating them through Apache Airflow's DockerOperator.
- Architected a medallion lakehouse on Databricks (Bronze/Silver/Gold) using PySpark and Delta Lake, improving data reliability by enforcing schema validation, deduplication, and derived feature engineering (home/away splits, win indicators) across every pipeline run.
- Resolved a critical data-loss bug causing systematic under-counts in production by redesigning the consumer's batch-flush logic and implementing idle-timeout polling, ensuring 100% record completeness (2,460/2,460) across every ingestion run.
- Orchestrated a fully automated 4-stage DAG (Kafka ingestion → Bronze → Silver → Gold) by integrating Airflow with Databricks' Jobs API via custom DatabricksSubmitRunOperator payloads, working around a provider-level limitation to enable serverless compute execution.
- Delivered stakeholder-facing insights through a Streamlit dashboard visualizing team-level home vs. away win percentages and performance splits across all 30 NBA teams, sourced directly from Databricks Gold-layer aggregations via the SQL Connector.

SKILLS

Languages: C, Python, JavaScript, PHP, Java, SQL

Web & APIs: HTML5, CSS3, JavaScript, REST API, JSON

Databases: PostgreSQL, MySQL, SQLite — normalized schema design

Cloud: AWS (EC2, S3, IAM, CloudWatch), Amazon Bedrock, OpenAI API **DevOps & tooling:** Docker, GitHub Actions, CI/CD pipelines